

Exercise 5 – Altimetry Techniques & Analysis

Answer Sheet

Several questions in this exercise involve interpreting images or other data generated in MATLAB. **Please save and include all such images/data as part of your responses.** When referring to a feature within an image, **be sure to clearly identify said feature** (e.g., with a box or arrow, or by specifying its image coordinates).

I. Geologic Background and Imaging Data

Q1. Describe the geologic setting and major geomorphological features you observe in the Narrow Angle image (*M0806185.gif*).

II. Photoclinometry

Q2. The solar azimuth is important, as only profiles that cut across the image parallel to the illumination direction can be used for deriving topography from photoclinometry. Explain briefly why this is the case.

Q3. Experiment with different albedos, letting the program guess first and then trying higher and lower values. Try this for a number of areas and try to figure out what goes wrong when the albedo is far away from the real answer. Explain it as best you can in words.

Do the same for the shadow illumination parameter. Export your relevant MATLAB plots when you're done, using File -> Export on the plot window.

III. Laser Altimetry

Q4. What is the total relief along this track?

Q5. Are all of the high points in this image at the same elevation?

What are the maximum slopes seen in this track, and are they consistent with loose sediment (as opposed to bedrock)?

(Note: Compare these slopes to the angle of repose: the steepest slope at which loose sediment is stable. The angle of repose observed for Martian dunes is about 30°.)

IV. Stereogrammetry

Q6. Notice that there are lots of areas where the automatic correlator couldn't find matches between the images. Why do you think this is?

(Note: This step requires installation of the Deep Learning Toolbox since `proftool3()` calls `minmax()`, which is a function in that toolbox.)

Q7. 1) Determine whether the layers exposed along the edge of this cliff are roughly horizontal or not.

2) Determine if the angle of the slope streaks which can be seen throughout the image is what you would expect for a fine dust layer on the surface (roughly 30° for loose sediment).

Q8. Type `hist(z(:),20)` to view a histogram of elevations for this DEM. What are the maximum and minimum elevations?